

N-ERGY Intern Take-Home Assignment

Overview

Build a mini document intelligence system that allows a user to:

- Upload PDFs
- Ask questions about them
- Receive answers with citations from the documents
- **(Bonus)** Suggest insights or next steps based on the documents

Requirements

Core Features

- Upload 1–50 PDFs
- Parse + process documents
- Ask natural language questions
- Return:
 - Answer
 - Cited sources (exact chunks or references)
- Simple working UI — function over form:
 - A browser-based interface (any framework or plain HTML)
 - Users should be able to upload PDFs, ask questions, and see answers + citations without touching the terminal
 - We are not evaluating design — we are evaluating whether you can wire a frontend to a backend cleanly

Technical Expectations

You are free to use any tools (including LLMs), but you must:

- Show a clear architecture
- Implement:
 - Document ingestion
 - Chunking strategy
 - Embeddings + retrieval
 - Answer generation
 - A functional frontend that connects to your backend
- Handle basic edge cases (e.g., empty PDFs, irrelevant queries)

Constraints

You must optimize for **ONE** of the following:

- **Latency** (fast responses)
- OR**
- **Accuracy** (higher quality answers)

Clearly state which you chose and why.

Development Environment

- You will complete this assignment on a shared development environment hosted by us.
- The environment includes access to Claude Code, which you may use if helpful.
- You are free to use any other tools or workflows you prefer.

Steps

Step 1: Send your SSH key

Email us your public SSH key (either as a .pub file or pasted in the email). We will use this to grant you access.

Step 2: Access the environment

Once access is granted, connect using:

```
ssh guestuser@ec2-3-239-244-156.compute-1.amazonaws.com
```

Step 3: Build your solution

- Build directly on the machine
- Install dependencies as needed
- Keep your work organized in a clearly named directory

Deliverables

Code

- Public GitHub repo
- Clean, runnable code

README (required)

Keep it concise but include:

- Architecture overview
- Key design decisions
- Tradeoffs you made
- What would break at scale (e.g., 10k+ documents)
- What you would improve with more time

Time Expectation

- **Target: 8 hours**
- We value clarity + decision-making over completeness

Submission

Send:

- GitHub repo link
- README